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United States Patent	12405512
Kind Code	B2
Date of Patent	September 02, 2025
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Blade driving device, camera device and portable electronic device

Abstract

Provided are a blade driving device, a camera device and a portable electronic device. The blade driving device includes a housing, a rotary body, a blade, at least one driving coil, a circuit board and at least one magnetic yoke. At least one rotating shaft extending into the accommodation space is disposed on the housing. The rotary body is a magnetic member with at least two magnetic poles. The blade includes one end fixedly connected to the rotary body and rotatable with the rotary body, and another end provided with a shielding portion and an opening portion. Compared with the related art, the present disclosure achieves compactness by omitting a number of mounted parts, realizing high efficient utilization of space and simplicity of structure and assembly, and enabling more efficient blade driving with a simpler structure.

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Appl. No.: **18/393584**

Filed: **December 21, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20250068035 A1	Feb. 27, 2025

Foreign Application Priority Data

JP	2023-134549	Aug. 22, 2023

Publication Classification

Int. Cl.: **G03B9/14** (20210101); **G03B30/00** (20210101)

U.S. Cl.:

CPC **G03B9/14** (20130101); G03B30/00 (20210101)

Field of Classification Search

CPC: G03B (9/04); G03B (9/08); G03B (9/14)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Japanese Patent Application No. 2023134549, entitled “BLADE DRIVING DEVICE, CAMERA DEVICE AND PORTABLE ELECTRONIC DEVICE,” filed Aug. 22, 2023, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present disclosure relates to the technical field of camera devices and, in particular, relates to a blade driving device, a camera device, and a portable electronic device.

BACKGROUND

(3) With the rapid development of camera technology, blade driving devices and lens drive devices have been widely used in many camera devices. Optical devices having blade driving devices are applied to various portable electronic device, such as mobile telephones and tablet computers, which are particularly popular with consumers.

(4) A blade driving device drives single or multiple blades to change a state of an opening, and is used as a shutter, an aperture, an aperture-compatible shutter, a filter or the like in various optical units such as cameras. In the blade driving device, the electromagnetic actuator is split at the assembly stage, and due to the interaction of deterioration of assemblability, increase in the number of parts, and loosening caused by the increase in the number of parts, the structural difficulty of the blade driving device itself may become high.

(5) Although there is a moving unit that uses a shape memory alloy with a small area in structure, a length of the shape memory alloy needs to be increased when a large amount of change is required.

(6) In addition, in the case where multiple arbitrary stopping positions are desired to be achieved with a single actuator, there is an issue that the difficulty of designing and manufacturing is increased due to the need for multi-pole magnetisation of a magnet, having to adjust the shape of a magnet sheet at a very tight angle, or the like.

(7) Therefore, it is desired to provide a novel blade driving device capable of solving the above issue.

EXISTING ART DOCUMENT

Patent Document

(8) Patent Document 1: Japanese Patent Application Laid-open No. 2014-10430
(9) Patent document 2: Japanese Patent Application Laid-open No. 2010-276654
(10) Patent Document 3: Japanese Patent Application Laid-open No. 10-68979
(11) Patent Document 4: Japanese Patent Application Laid-open No. 2019-203944
(12) Patent Document 5: Japanese Patent Application Laid-open No. 2019-148699
(13) Patent Document 6: Japanese Patent Application Laid-open No. 2018-162621.

SUMMARY

(14) The present disclosure provides a blade driving device, a camera device, and a portable electronic device, in order to solve the issue of increasing in difficulty of designing and manufacturing the structure of a blade driving device with which a user appropriately adjusts the amount of light in shooting.

(15) In one aspect, a blade driving device is provided. The blade driving device includes a housing, a rotary body, a blade, at least one driving coil, a circuit board and at least one magnetic yoke.

(16) The housing includes a base and a cover plate, where the base has a first through-hole, the cover plate has a second through-hole, the first through-hole and the second through-hole form an optical axis channel, the base and the cover plate form an accommodation space, and at least one rotating shaft extending into the accommodation space is disposed on the housing.

(17) The rotary body is a magnetic member with at least two magnetic poles, provided with a pivot hole sleeved on a respective rotating shaft of the at least one rotating shaft, and rotatable relative to the respective rotating shaft.

(18) The blade has one end fixedly connected to the rotary body and rotatable with the rotary body, and another end provided with a shielding portion and an opening portion, where the shielding portion is formed in a circumferential direction of the opening portion, and the optical axis channel is located in a moving path of the opening portion.

(19) An orthographic projection of the rotary body in a direction of the optical axis channel has an overlapping area with the at least one driving coil.

(20) A circuit board is electrically connected to the at least one driving coil.

(21) At least one magnetic yoke, fixed to the housing.

(22) As an improvement, the rotary body, the at least one driving coil and the circuit board are disposed sequentially along the optical axis channel, two magnetic yokes spaced apart from each other and disposed at an included angle with each other are provided, and orthographic projections of the two magnetic yokes have overlapping areas with the rotary body.

(23) As an improvement, the two magnetic yokes are located on a side of the at least one driving coil away from the rotary body.

(24) As an improvement, two driving coils are provided, and disposed opposite to different magnetic poles of the rotary body, respectively.

(25) As an improvement, two blades and two rotary bodies are disposed in the accommodation space, each of the two blades are fixed to a respective rotary body of the two rotary bodies, the two blades have opening portions with different inner diameters, and the optical axis channel is located within moving paths of the two blades.

(26) As an improvement, the blade includes a plurality of opening portions, and the optical axis channel is located within a moving path of each of the plurality of opening portions.

(27) As an improvement, the plurality of opening portions have different inner diameters.

(28) As an improvement, a position detecting element configured to detect a magnetic flux of the rotary body is disposed in the accommodation space.

(29) As an improvement, the blade is provided with a countersink hole, and the rotary body extends

into the countersink hole to form a rigid connection.
(30) In a second aspect, a camera device is provided. The camera device includes the preceding blade driving device
(31) In a third aspect, a portable electronic device is provided. The portable electronic device includes the preceding camera device.
(32) A portable electronic device, comprising the camera device according to any one of claim **10**.
(33) Compared with the related art, in the present disclosure, one end of the blade is connected to the rotary body to form a one-piece structure and the blade is rotatable with the rotation of the rotary body, thereby overcoming the technical problems of difficult assembly and increased structure difficulty caused by the split driving structure in the related art. In addition, compactness is achieved by omitting a number of mounted parts, realizing high efficient utilization of space and simplicity of structure and assembly, and enabling more efficient blade driving with a simpler structure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a front view of a blade driving device according to Embodiment one of the present disclosure.
(2) FIG. **2** is a sectional view taken along line A-A of FIG. **1**.
(3) FIG. **3** is a side view of the blade driving device according to Embodiment one of the present disclosure.
(4) FIG. **4** is a rear view of the blade driving device according to Embodiment one of the present disclosure.
(5) FIG. **5** is an exploded perspective view of the blade driving device according to Embodiment one of the present disclosure.
(6) FIG. **6** is a front view of a blade of a first structure of the blade driving device according to Embodiment one of the present disclosure.
(7) FIG. **7** is a front view of a blade of a second structure of the blade driving device according to Embodiment one of the present disclosure.
(8) FIG. **8** is a schematic diagram illustrating structures of a rotary body and a driving coil in a first mating state of the blade driving device according to Embodiment one of the present disclosure.
(9) FIG. **9** is a schematic diagram illustrating structures of a rotary body and a driving coil in a second mating state of the blade driving device according to Embodiment one of the present disclosure.
(10) FIG. **10** is a schematic diagram illustrating structures of a rotary body and a driving coil in a third mating state of the blade driving device according to Embodiment one of the present disclosure.
(11) FIG. **11** is a schematic diagram illustrating structures of a rotary body and a magnetic yoke in a first mating state of the blade driving device according to Embodiment one of the present disclosure.
(12) FIG. **12** is a schematic diagram illustrating structures of a rotary body and a magnetic yoke in a second mating state of the blade driving device according to Embodiment one of the present disclosure.
(13) FIG. **13** is an exploded perspective view of a blade driving device according to Embodiment two of the present disclosure.
(14) FIG. **14** is a schematic diagram illustrating a structure of a portable electronic device including the blade driving device of the present disclosure.
REFERENCE LIST

(15) TABLE-US-00001 10 housing 11 base 12 cover plate 13 accommodation space 14 rotating shaft 15 first through hole 16 second through hole 17 limiting block 20 blade 21 shielding 22 opening portion portion 23 countersink hole 30 rotary body 31 pivot hole 40 driving coil 50 circuit board 60 magnetic yoke 100 blade driving device 200 camera device 300 portable electronic device

DETAILED DESCRIPTION OF THE EMBODIMENTS

(16) The embodiments described below with reference to the drawings are merely exemplary and intended to explain the present disclosure, and are not to be construed as limiting the present disclosure.

Embodiment One

(17) As shown in FIGS. **1** to **12**, an embodiment of the present disclosure provides a blade driving device **100** applied in a camera device **200**, so as to control the amount of light entering the camera device **200** by blocking unwanted light on an optical axis channel with a blade **20**.

(18) In the embodiment provided in the present disclosure, the blade driving device **100** includes a housing **10**, a rotary body **30**, and a blade **20**.

(19) The housing **10** is internally provided with an accommodation space **13** in which an optical axis channel is formed. The housing **10** is provided with a first through hole **15** and a second through hole **16**, which are coaxially arranged to form the optical axis channel.

(20) In a feasible implementation, referring to FIG. **2** and FIG. **5**, the housing **10** is of a split structure to facilitate molding and assembly. Specifically, the housing **10** includes a base **11** and a cover plate **12**, the first through hole **15** is disposed on the base **11**, the second through hole **16** is disposed on the cover plate **12**, the base **11** is of a box structure with one end open, and the cover plate **12** covers the open end by a fixing manner such as clamping, bonding or bolting to form the accommodation space **13**.

(21) Referring to FIGS. **2** and **5**, a rotating shaft **14** is disposed in the accommodation space **13**. In a feasible implementation, the rotating shaft **14** is integrally formed with the base **11** to facilitate molding processing. The rotating shaft **14** is a cylindrical shaft body and is disposed adjacent to the optical axis channel.

(22) The rotary body **30** is provided with a pivot hole **31**, and the rotating shaft **14** is clearance fitted in the pivot hole **31**, so that the rotary body **30** is rotatable with an axial direction of the rotating shaft **14** as a rotation axis. During assembly, it is only necessary to open the cover plate **12** and bring the pivot hole **31** into alignment with the rotating shaft **14** for installation, thus reducing the manufacturing difficulty. The rotating shaft **1** is clearance fitted in the pivot hole **31**, and the thickness of the rotating shaft **14** is coincided with the thickness of the rotary body **30**, thereby reducing the space occupied by the housing **10** in a direction of the optical axis channel and facilitating the miniaturization of the blade driving device **100**.

(23) The blade **20** is disposed in the accommodation space **13**, and has a first end connected to the rotary body **30**. In a feasible implementation, the first end of the blade **20** is provided with a stepped countersink hole **23**, the rotary body **30** extends into the countersink hole **23** and is fixed in the countersink hole **23** by a fixing manner such as clamping, bonding or bolting. The blade **20** is rotatable synchronously with the rotation of the rotary body **30**. By extending the rotary body **30** into the countersink hole **23**, the space occupied by the housing **10** in the direction of the optical axis channel can be further compressed, and the countersink hole **23** can also play a role in limiting the rotary body **30**. The shape and size of the countersink hole **23** may be determined according to the shape and size of the rotary body **30**, which is not limited herein.

(24) Since the blade is made in a very thin shape, in the related art, a large amount of friction locally concentrates on a shaft member for supporting the blade due to repeated rotation of the blade, resulting in the generation of pollutants from the shaft member and the damage to the shaft member itself. In the present disclosure, by rigidly connecting the rotary body **30** to the blade **20**, a large area of contact is maintained between the pivot hole **31** of the rotary body **30** and the rotating shaft **14**, so that local stress concentration is not generated on the shaft member, and the durability

can be greatly improved.
(25) Referring to FIG. **5** and FIG. **6**, a shielding portion **21** and an opening portion **22** are formed on a second end of the blade **20**. The shielding portion **21** is formed in a circumferential direction of the opening portion **22**, and the optical axis channel is located in a moving path of the opening portion **22**. When the blade **20** is moved to a preset position, the opening portion **22** is coaxial with the optical axis channel. The opening portion **22** and the shielding portion **21** of the blade **20** can be configured in any shape which enables optically optimal light-shielding shape to achieve the purpose of improving performance.
(26) Referring to FIG. **2** and FIG. **6**, the opening portion **22** is of a through-slot structure penetrating through the blade **20**. In this embodiment, the opening portion **22** is a circular slot. As appreciated by those skilled in the art, the opening portion **22** may be in other shape, such as a polygon of a triangle or more (e.g., a regular hexagon), which is not limited herein. When the blade **20** is rotated to a preset position, an axis of the first through-hole **15**, an axis of the opening portion **22** and an axis of the second through-hole **16** coincide with each other. By changing an inner diameter or a shape of the opening portion **22**, an amount of light entering the camera device **200** through the optical axis channel can be adjusted.
(27) The shielding portion **21** is located in the circumferential direction of the opening portion **22**, and is configured to block the passage of light. A blade **20** optimally matched with a light source may be appropriately used. In the case of visible light, a shielding portion **21** made of plastic coated in black may be selected. For light other than visible light, a shielding portion **21** made of metal may also be used, so as to meet the requirement of blocking the passage of light.
(28) Further, in order to drive the rotary body **30** to rotate, in embodiments of the present disclosure, the rotary body **30** is a magnetic member and magnetized into a single-layer magnet having an N pole and an S pole, and a driving coil **40** and a circuit board **50** are disposed in the accommodation space **13**. In a feasible implementation, the rotary body is a cylinder, the N pole and the S pole are disposed symmetrically along a radial direction of the rotating shaft **14**, and a half of the rotary body by every 180° is a magnetic pole. As appreciated by those skilled in the art, one magnetic pole may be arranged by every 90°, and then the rotary body has four magnetic poles. In this case, two driving coils **40** may be disposed in parallel, or in an “L”-like shape.
(29) The rotary body **30**, the driving coil **40** and the circuit board **50** are disposed in sequence along the optical axis channel. The driving coil **40** is electrically connected to the circuit board **50**. The rotary body **30** is driven to rotate by an electromagnetic force through the energization of the driving coil **40**, and the blade **20** can be rotated together with the rotary body **30** to a preset position. At the preset position, the opening portion **22** forms a concentric hole structure with the first through hole **15** and the second through hole **16**.
(30) The driving principle of the rotary body **30** is as follows.
(31) When the driving coil **40** is energized, a Lorentz force is generated in the driving coil **40** by the interaction between a magnetic field of the rotary body **30** and a current flowing in the driving coil **40**. Since the driving coil **40** is fixed, a reaction force acts on the rotary body **30** and becomes a driving force of the rotary body **30**, so that the rotary body **30** is rotated with an axis direction of the rotating shaft **14** as a center.
(32) A rotating direction of the rotary body **30** may be changed by changing a direction of the current flowing in the driving coil **40**, so that the rotary body **30** can be rotated clockwise or counterclockwise in a plane orthogonal to the optical axis channel.
(33) In a feasible implementation, as shown in FIG. **8**, two driving coils **40** are provided, the rotary body **30** is disposed between the two driving coils **40**, and an orthographic projection of the rotary body **30** in the direction of the optical axis channel falls on the two driving coils **40**, so that the rotary body **30** can utilize magnetic forces of the two driving coils **40** at the same time, whereby the rotary body **30** can be subjected to a greater driving force, thereby improving the rotation efficiency of the blade **20**.

(34) In another feasible implementation, as shown in FIG. **9**, only one driving coil **40** is provided, the driving coil **40** includes a left half portion and a right half portion symmetrically arranged along a length direction of the driving coil **40**, and an orthographic projection of the rotary body **30** in the direction of the optical axis channel falls on both the left half portion and the right half portion, which helps to improve the simplicity of assembly and save space on the premise of satisfying the driving force.
(35) In another feasible implementation, as shown in FIG. **10**, only one driving coil **40** is provided, the driving coil **40** includes a left half portion and a right half portion symmetrically arranged along a length direction of the driving coil **40**, and an orthographic projection of the rotary body **30** in the direction of the optical axis channel falls on either the left half portion or the right half portion, which helps to improve the simplicity of assembly and save space on the premise of satisfying the driving force.
(36) In the embodiments provided in the present disclosure, in order to keep the rotary body **30** at a preset position, a magnetic yoke **60**, as a magnet piece, is disposed at the boundary of the magnetic poles of the rotary body **30** at an appropriate rotating position of the blade **20**. The magnetic yoke **60** is disposed at a preset position, so that during the rotation of the blade **20**, when the boundary of the magnetic poles of the rotary body **30** comes into alignment with the magnetic yoke **60**, the blade **20** stops rotating, and the opening portion **22** is coaxial with the optical axis channel. The magnetic yoke **60** is located on a side of the driving coil **40** away from the rotary body **30**, and an orthographic projection of the magnetic yoke **60** in the direction of the optical axis channel has an overlapping area with the rotary body **30**. The magnetic yoke **60**, which is a magnet piece, always applies a force attracting the rotary body **30** to the base **11**, and serves to keep a distance between the rotary body **30** and the driving coil **40** constant while the blade **20** is driven by an electromagnetic force.
(37) Further, a plurality of magnetic yokes **60** are provided, and orthographic projections of the magnetic yokes **60** in the direction of the optical axis channel fall on a moving path of the blade **20** so that the blade **20** has a plurality of stop positions. Referring to FIG. **11** and FIG. **12**, in FIG. **11**, there are two magnetic yokes **60**, which are intersected to form a "V"-like shape or an "L"-like shape; and in FIG. **12**, there are three magnetic yokes **60**, which are intersected to form a "Y"-like shape. The plurality of magnetic yokes **60** are annularly spaced in the axial direction of the rotary body **30**. Magnetic yokes **60** provided by the base **11** may be configured in switching positions of the two points as shown in FIG. **1**, in switching positions of the three points as shown in FIG. **12**, or other switching positions, as long as the magnetic yokes **60** are configured corresponding to positions where the magnetic poles of the rotary body **30** are desired to be stopped, to be able to cause the blade **20** having the opening portion **22** with an appropriate diameter to stop at a plurality of predetermined positions.
(38) Further, as shown in FIG. **5**, the base **11** is protruded with a limiting block **17** configured to limit a range of a moving path of the rotary body **30**. When the rotary body **30** is rotated to abut against the limiting block **17**, the rotary body **30** reaches the end of the moving path and cannot continue to rotate.
(39) Referring to FIG. **7**, one blade **20** is provided with a plurality of opening portions **22**, and the optical axis channel is located in a moving path of each of the plurality of opening portions **22**. The blade **20** has a plurality of preset positions. At each preset position, a respective opening portion **22** of the plurality of opening portions **22** corresponding to the optical axis channel is provided so that the axial directions of the first through hole **15**, the respective opening portion **22** and the second through hole **16** coincide with each other. Further, each opening portion **22** is different in inner diameter, that is, each opening portion **22** has a different aperture. By providing a plurality of apertures, a plurality of shapes of the opening portions **22** can be appropriately selected for even a single blade **20**, thereby forming a blade **20** with good optical characteristics.
(40) In the embodiments provided by the present disclosure, a position detecting element (not

shown) capable of detecting a magnetic flux of the rotary body **30** is disposed in the accommodation space **13**. By detecting the magnetic flux of the rotary body **30**, a position of the rotary body **30** can be accurately detected, and then a position of the blade **20** can be determined. The position detecting element may be disposed in the driving coil **40** and located in the same plane as the driving coil **40**, thus further improving the space utilization rate and reducing a height of the housing **10**.

Embodiment Two

(41) In this embodiment, different from Embodiment one, a plurality of blades **20** may be disposed in the accommodation space **13**, the optical axis channel is located in moving paths of the plurality of blades **20**. Correspondingly, a plurality of pivot shafts are disposed in the accommodation space **13**, one rotary body **30** is provided on each of the plurality of pivot shafts, each of the plurality of blades **20** is correspondingly fixed on a respective rotary body **30**, and a driving coil **40** and a magnetic yoke **60** are provided for each of the plurality of rotary bodies **30**, thus ensuring the degrees of freedom matched with the blades **20**.

(42) Specifically, as shown in FIG. **13**, two blades **20** are disposed in the accommodation space **13**, and located on two opposite sides of the optical axis channel respectively, the optical axis channel is located in moving paths of the two blades **20**, and a driving coil **40** and a magnetic yoke **60** are correspondingly provided for each of two rotary bodies **30**, and the inner diameters of the opening portions **22** provided on the two blades **20** may be the same or different.

(43) On the basis of the blade driving device **100** provided in the above embodiments, the present disclosure further provides a camera device **200** including the preceding blade driving device **100**. In the blade driving device **100**, the rotary body **30** driven by electromagnetism is integrated with the blade **4**, which features excellent assembly performance. The number of components such as a driving coil **40** and a magnetic yoke **60** can be reduced as required, and a plurality of arbitrary stop positions can be realized by one rotary body **30**, which also greatly reducing the design difficulty. In combination with the light amount adjustment provided by the blade driving device, the quality of the photographed image can be improved.

(44) Further referring to FIG. **14**, the present disclosure further provides a portable electronic device **300**, such as a smart phone or a tablet device, including the preceding camera device **200**.

(45) The structures, features and effects of the present disclosure are described above in detail in accordance with the embodiments shown in the drawings. The above are merely preferred embodiments of the present disclosure, and not intended to limit the scope of the present disclosure. Any changes made in accordance with the concept of the present disclosure, or equivalent embodiments modified by equivalent changes, are not beyond the spirit covered by the specification and drawings, and shall be within the scope of protection of the present disclosure.

Claims

1. A blade driving device, comprising: a housing, including a base and a cover plate, wherein the base has a first through-hole, the cover plate has a second through-hole, the first through-hole and the second through-hole form an optical axis channel, the base and the cover plate form an accommodation space, and the housing is provided with at least one rotating shaft extending into the accommodation space; a rotary body, being a magnetic member with at least two magnetic poles, provided with a pivot hole sleeved on a respective rotating shaft of the at least one rotating shaft, and rotatable relative to the respective rotating shaft; a blade, having one end fixedly connected to the rotary body and rotatable with the rotary body, and another end provided with a shielding portion and an opening portion, wherein the shielding portion is formed in a circumferential direction of the opening portion, and the optical axis channel is located in a moving path of the opening portion; at least one driving coil, wherein an orthographic projection of the rotary body in a direction of the optical axis channel has an overlapping area with the at least one

driving coil; a circuit board, electrically connected to the at least one driving coil; and at least one magnetic yoke, fixed to the housing; wherein the blade is provided with a countersink hole, and the rotary body extends into the countersink hole to form a rigid connection.
2. The blade driving device according to claim 1, wherein the rotary body, the at least one driving coil and the circuit board are disposed sequentially along the optical axis channel, two magnetic yokes spaced apart from each other and disposed at an included angle with each other are provided, and orthographic projections of the two magnetic yokes have overlapping areas with the rotary body.
3. The blade driving device according to claim 2, wherein the two magnetic yokes are located on a side of the at least one driving coil away from the rotary body.
4. The blade driving device according to claim 3, wherein two driving coils are provided, and disposed opposite to different magnetic poles of the rotary body, respectively.
5. The blade driving device according to claim 4, wherein two blades and two rotary bodies are disposed in the accommodation space, each of the two blades are fixed to a respective rotary body of the two rotary bodies, the two blades have opening portions with different inner diameters, and the optical axis channel is located within moving paths of the two blades.
6. The blade driving device according to claim 1, wherein the blade includes a plurality of opening portions, and the optical axis channel is located within a moving path of each of the plurality of opening portions.
7. The blade driving device according to claim 6, wherein the plurality of opening portions have different inner diameters.
8. The blade driving device according to claim 2, wherein a position detecting element configured to detect a magnetic flux of the rotary body is disposed in the accommodation space.
9. A camera device, comprising: a blade driving device, wherein the blade driving device includes: a housing, including a base and a cover plate, wherein the base has a first through-hole, the cover plate has a second through-hole, the first through-hole and the second through-hole form an optical axis channel, the base and the cover plate form an accommodation space, and the housing is provided with at least one rotating shaft extending into the accommodation space; a rotary body, being a magnetic member with at least two magnetic poles, provided with a pivot hole sleeved on a respective rotating shaft of the at least one rotating shaft, and rotatable relative to the respective rotating shaft; a blade, having one end fixedly connected to the rotary body and rotatable with the rotary body, and another end provided with a shielding portion and an opening portion, wherein the shielding portion is formed in a circumferential direction of the opening portion, and the optical axis channel is located in a moving path of the opening portion; at least one driving coil, wherein an orthographic projection of the rotary body in a direction of the optical axis channel has an overlapping area with the at least one driving coil; a circuit board, electrically connected to the at least one driving coil; and at least one magnetic yoke, fixed to the housing; wherein the blade is provided with a countersink hole, and the rotary body extends into the countersink hole to form a rigid connection.
10. The camera device according to claim 9, wherein the rotary body, the at least one driving coil and the circuit board are disposed sequentially along the optical axis channel, two magnetic yokes spaced apart from each other and disposed at an included angle with each other are provided, and orthographic projections of the two magnetic yokes have overlapping areas with the rotary body.
11. The camera device according to claim 10, wherein the two magnetic yokes are located on a side of the at least one driving coil away from the rotary body.
12. The camera device according to claim 11, wherein two driving coils are provided, and disposed opposite to different magnetic poles of the rotary body, respectively.
13. The camera device according to claim 9, wherein two blades and two rotary bodies are disposed in the accommodation space, each of the two blades are fixed to a respective rotary body of the two rotary bodies, the two blades have opening portions with different inner diameters, and the optical

axis channel is located within moving paths of the two blades.
14. The camera device according to claim 9, wherein the blade includes a plurality of opening portions, and the optical axis channel is located within a moving path of each of the plurality of opening portions.
15. The camera device according to claim 14, wherein the plurality of opening portions have different inner diameters.
16. The camera device according to claim 10, wherein a position detecting element configured to detect a magnetic flux of the rotary body is disposed in the accommodation space.
17. A portable electronic device, comprising: a camera device, wherein the camera device includes: a blade driving device, wherein the blade driving device includes: a housing, including a base and a cover plate, wherein the base has a first through-hole, the cover plate has a second through-hole, the first through-hole and the second through-hole form an optical axis channel, the base and the cover plate form an accommodation space, and the housing is provided with at least one rotating shaft extending into the accommodation space; a rotary body, being a magnetic member with at least two magnetic poles, provided with a pivot hole sleeved on a respective rotating shaft of the at least one rotating shaft, and rotatable relative to the respective rotating shaft; a blade, having one end fixedly connected to the rotary body and rotatable with the rotary body, and another end provided with a shielding portion and an opening portion, wherein the shielding portion is formed in a circumferential direction of the opening portion, and the optical axis channel is located in a moving path of the opening portion; at least one driving coil, wherein an orthographic projection of the rotary body in a direction of the optical axis channel has an overlapping area with the at least one driving coil; a circuit board, electrically connected to the at least one driving coil; and at least one magnetic yoke, fixed to the housing; wherein the blade is provided with a countersink hole, and the rotary body extends into the countersink hole to form a rigid connection.
18. The portable electronic device according to claim 17, wherein the rotary body, the at least one driving coil and the circuit board are disposed sequentially along the optical axis channel, two magnetic yokes spaced apart from each other and disposed at an included angle with each other are provided, and orthographic projections of the two magnetic yokes have overlapping areas with the rotary body.